Loss Functions and Optimization

Scaler School of Technology | Deep Learning I | Week 2 — Session 3/3 | 9 May 2026

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What Is This Lecture About?

Session 3 gave us backpropagation — the algorithm that computes gradients for every weight in a network. Today we answer the two remaining questions that complete the training loop: *what* should we minimize? (loss functions) and *how* should we use the gradients? (optimizers). By the end of this session, you will have a complete mental model of neural network training.

1 Loss Functions: MSE & Cross-Entropy

The loss function measures how wrong a prediction is. MSE squares the error — natural for regression. Cross-entropy uses logarithmic penalties — devastating for confident wrong answers in classification. *Think of it as the grading rubric: it decides how harshly mistakes are punished.*

2 Gradient Descent Variants & Momentum

Once we have gradients, how do we use them? Batch GD uses all data (slow but stable), SGD uses one example (fast but noisy), mini-batch is the sweet spot. Momentum adds velocity — like a ball rolling downhill instead of stopping at every step. *Think of it as the difference between walking and rolling.*

3 Adaptive Methods & Learning Rate Schedules

One learning rate for all parameters is like one shoe size for all feet. AdaGrad, RMSProp, and Adam adapt per-parameter. Adam is the default for 90% of problems. Learning rate schedules (cosine annealing, warmup) improve convergence further. *Think of it as an autopilot that adjusts speed for each stretch of road.*

What to Expect in the Session

- Walk through MSE and cross-entropy with worked gradient examples
- See why MSE + sigmoid causes vanishing gradients (and how cross-entropy fixes it)
- Compare batch, stochastic, and mini-batch gradient descent on contour plots
- ⋆ Watch SGD, Momentum, and Adam race on the same loss surface
- Build intuition for Adam's bias correction with a step-by-step example
- Learn a practical recipe: loss + optimizer + schedule for any project

How to Prepare

✓ **Recall** the chain rule and backpropagation from Session 3 — we build directly on it

✓ **Brush up on** logarithms: $\log(0.01) \approx -4.6$, $\log(0.99) \approx -0.01$ — cross-entropy lives in log-space

✓ **Think about:** "If I have the gradient, why not just subtract it from the weights? What could go wrong?"

✓ **Optional:** Skim the PyTorch `torch.optim` docs — see how many optimizers exist

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Duration: ~2 hours — Prerequisites: Sessions 1–3 (forward pass, backpropagation) — Instructor: Priyansh Saxena